FX-Driven Debt Erosion in Local-Currency DeFi Lending: Counterfactual Evidence from MakerDAO ETH-A Draws

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Abstract

We use 130,742 MakerDAO ETH-A debt draws as a source of observed principal sizes and timestamps for hypothetical Argentine-peso (ARS) and Turkish-lira (TRY) loans, not as a representative sample of local-currency borrowers. The 7,105 selected clean, repaid spells with exchange-rate coverage last a median 6.95 days. Matching their endpoints to published daily official reference rates gives median counterfactual net benefits of **−2.86**% (ARS) and **−4.30**% (TRY) under an illustrative 20% effective annual borrowing rate and specified costs. Monthly OECD/FRED marks give **−3.00**% and **−4.14**% for the same spells and mechanically suppress same-month FX movement. The corresponding 12-month medians, 2.67% and 6.59%, impose a holding period on all eligible draws and are not observed borrowing outcomes. Borrowing-rate and position-size sensitivities, repeated draws by urn, and concentration among large positions limit generalisation. Daily published reference quotes are retrospective marks, not demonstrated executable conversion prices, especially for ARS. Conditional collateral-sale accounting is kept separate from the empirical repayment comparisons; neither liquidation frequency nor protocol solvency is estimated.

Keywords: Decentralized finance, foreign exchange, local-currency borrowing, MakerDAO, borrower cash flow, liquidation

1 Introduction

A borrower who owes a fixed number of local-currency units (LCU) can face a smaller USD-equivalent claim after that currency depreciates. This is a foreign-exchange accounting effect, not a measurement of consumer-price inflation. Whether it survives financing costs depends on the loan duration, position size, feasible exchange rates, and collateral outcome. A MakerDAO ETH-A draw is a DAI-denominated transaction, not evidence that anyone borrowed ARS or TRY stablecoins.

We place short reconstructed loan spells before long holding-period scenarios and distinguish voluntary repayment from nonrecourse collateral close-out. A liquidated borrower does not also repay the debt at the planned horizon. The empirical question is what observed draw sizes and timestamps imply under specified *hypothetical* local-currency debt and voluntary-repayment assumptions. The collateral analysis asks only whether a terminal mark would fall below an assumed threshold. It cannot identify an actual liquidation time, auction result, or borrower return in that branch.

The contribution is an auditable mapping from public on-chain draws and official FX rates to conditional borrower-side accounting, with daily marks for short spells and robustness to borrowing rates, repeated draws, small positions, duration, and concentration. Collateral-sale and balance-sheet identities delimit the analysis rather than provide empirical solvency estimates. Prior work on lending and

liquidation motivates the distinction between a collateral threshold and an executed sale [1, 2, 13]. Stablecoin research separates redemption and arbitrage from the value of an individual loan [3–5].

2 Related research and scope

The relevant economic mechanism is a change in the dollar value of a fixed *nominal foreign-currency liability*. It need not coincide with consumer-price inflation: an ARS or TRY price index and an ARS/USD or TRY/USD exchange rate answer different questions. A borrower who retains dollar-valued proceeds while owing a fixed number of LCU units may experience an accounting reduction in dollar debt after depreciation. Whether that accounting effect can be obtained in a market depends on an accessible conversion rate, the cost of acquiring the repayment token and the time for which the loan remains open. This is why we compare selected observed loan durations with imposed longer horizons and refrain from describing an official FX mark as an executable trade.

Research on DeFi lending describes smart-contract credit and its dependence on collateral and oracle rules [1, 2]. Empirical liquidation research illustrates why a marked collateral threshold should not be equated with an executed auction at that price [13]. A collateral sale changes ownership of the posted asset and settles as much of the debt as its proceeds permit. Treating that sale and full voluntary repayment as simultaneous costs would misstate both borrower and issuer cash flows. Our conditional ledger makes the two outcomes mutually exclusive, but the trace and FX files used here cannot identify the prices and timing needed to estimate how often either branch would occur in a hypothetical LCU vault.

Stablecoin prices additionally depend on redemption arrangements, arbitrage access and the structure of backing [3–5]. Those system-wide outcomes are not measured by an individual borrower's change in the dollar value of a liability. The research gap addressed here is therefore narrower than a test of a new protocol: we link public draw dates and amounts to official currency histories and explicit cost assumptions, then check whether the resulting descriptions survive changes in observational unit, duration and position size. The proposed balance-sheet identity specifies what additional data a solvency study would require.

3 Data and observational units

3.1 Draw events, urns, and clean spells

The public replication archive accompanying Chaleenutthawut et al. [6] contains 4,466,880 successful MakerDAO Vat traces. Using the documented debt-change calculation for `frob` [7], we retain 130,742 positive ETH-A draws of at least 1 DAI between January 2020 and July 2023, totalling 13.318 billion DAI, in 16,846 urn identifiers. The trace archive's exclusion of top-level calls limits the completeness of inferred debt histories. A draw is an event, not an independent borrower or necessarily a complete lifecycle; an urn is a vault identifier, not a verified person. The selection rules and archive checksum are in the accompanying documentation.

Table 1 Auditable sample construction. Counts reproduce `data/processed/sample_construction.csv`; successive exclusions apply after the positive ETH-A draw selection. Clean spells are a separate, restricted subset rather than all 130,742 draw events.

Stage	Count
Successful Vat call traces in public source archive	4,466,880
Decoded ETH-A debt-changing events	229,224
Positive ETH-A `frob` debt draws	137,426
Excluded before 1 January 2020	6,527
Excluded below 1 DAI after date filter	157
Final positive draw events	130,742
Distinct urn identifiers in final events	16,846
Reconstructed clean, repaid spells before FX coverage	7,947

The resulting debt-change distribution is highly skewed: the median is 4,792 DAI, the mean 101,865 DAI and the maximum 150.45 million DAI. These values are independently recoverable from the tracked processed event file. The transaction hash and urn remain attached to each event for auditing. A positive `dart` can represent a further draw from an existing vault; neither the number of rows nor the number

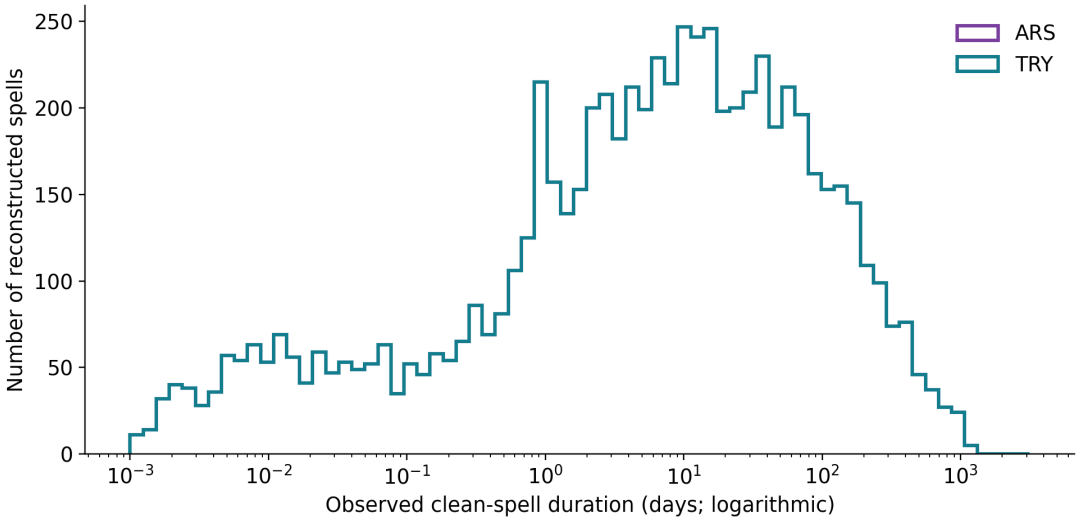


Fig. 1 Duration of 7,105 restricted clean, repaid spells with FX coverage per currency. The same selected spells are mapped to ARS and TRY; the horizontal axis is logarithmic. This subset is not a survival distribution for all draws.

of urns identifies the number of people. In particular, a draw does not establish the time at which its associated debt was ultimately repaid.

MakerDAO is useful here because ETH-A debt draws provide recorded principal changes, UTC event times and urn identifiers under a transparent on-chain collateralised-debt mechanism. We map a DAI draw's USD-equivalent size and calendar exposure to a *stipulated* LCU liability, then vary the hypothetical closing date; this holds the size and timing inputs fixed while exposing the effect of an FX path and costs. It does not follow that ARS or TRY borrowers would choose those loan sizes or entry months, face the same collateral constraints, or have access to equivalent liquidity. These data have one asset type and one historical adoption window, repeated activity within urns, skewed sizes and selected reconstructible closures. Thus neither the event distribution nor the clean-spell duration distribution represents an LCU credit market. We report size floors, urn summaries and month concentration to make these constraints visible, not to claim external validity.

We reconstruct a restricted subset of single-draw, single-repayment, non-forked clean spells. There are 7,105 spells with both monthly and daily FX coverage for each currency. Their median observed duration is 6.95 days. These are selected repaid spells and cannot estimate the duration or liquidation frequency of all vaults. Observed timestamps determine duration, while all ARS/TRY cash flows remain *counterfactual*; MakerDAO did not issue an LCU loan.

3.2 Published exchange-rate references

E_t denotes LCU per USD. The fixed-horizon scenarios retain archived OECD monthly-average series `ARGCCUSMA02STM` and `CCUSMA02TRM618N` distributed through FRED [9, 10]. Using these same monthly series for a seven-day spell is problematic: entry and exit in one calendar month share one rate and gross FX benefit becomes exactly zero. For the observed-duration analysis we therefore use archived daily published references: the BCRA Communication A3500 wholesale ARS/USD reference [11] and the ECB's TRY/EUR rate divided by its USD/EUR rate to obtain TRY/USD [12]. For each UTC event date the code takes the most recent published reference *calendar date no later than that date*, without interpolation or a future date. On nonpublication days it carries forward the preceding quote. The median quote-date lag is zero, its 95th percentile two days and its maximum four days across the selected endpoints. The same selected spells are also valued at arithmetic monthly means of their corresponding daily source and at the original OECD/FRED monthly marks to distinguish a change of frequency from a change of source.

These marks are retrospective reference valuations; a published quote on the same date could have appeared after the on-chain event, and the data do not show a simultaneous tradable rate. BCRA's wholesale reference is not a demonstrated retail conversion price. An Argentine rate accessible under exchange controls can differ materially from any official series. No independently vetted parallel-market series is included, so no numerical estimate is labelled executable; daily frequency fixes the zero-within-month artefact but cannot resolve market access or intraday execution.

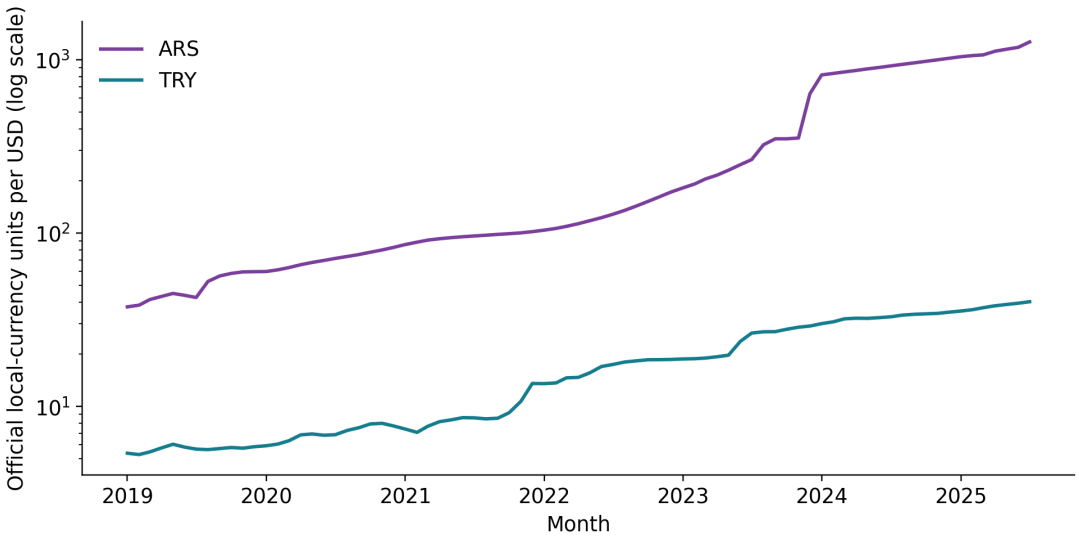


Fig. 2 Archived official monthly-average exchange rates, in local-currency units per US dollar on a logarithmic vertical axis. The series do not establish a freely executable conversion price.

If an executable ARS rate were $E_t^{\text{exec}} = w_t E_t^{\text{official}}$ for a premium $w_t > 0$, the gross effect would be $1 - (w_0/w_h)(E_0^{\text{official}}/E_h^{\text{official}})$. An unknown change in the premium can reverse even the sign of an official-rate benefit. This identity is a sensitivity argument, not an estimated parallel-rate path.

4 Conditional borrower cash-flow model

4.1 Voluntary repayment

For a draw of P_i USD-equivalent DAI at origination, hypothetical initial LCU debt is $L_{i,0} = P_i E_{i,0}$. With an *effective annual* LCU borrowing rate r, and no intermediate payments, its USD value at voluntary closing after t years is

$$D_i(t) = P_i \frac{E_{i,0}}{E_{i,t}}(1+r)^t, \qquad G_i(t) = P_i - D_i(t)\big|_{r=0}. \tag{1}$$

The protocol fee fP_i is paid externally in USD at opening and is *not financed*. We assume a conversion friction sP_i on the opening swap of hypothetical LCU proceeds into USD and $sD_i(t)$ on acquiring LCU for voluntary repayment. The illustrative gas budget g is split equally: $g/2$ at opening and $g/2$ for voluntary closing. The incremental USD cash-accounting benefit relative to holding the same collateral without borrowing is

$$B_i^{\text{repay}}(t) = P_i - fP_i - sP_i - \tfrac{g}{2} - D_i(t) - sD_i(t) - \tfrac{g}{2}. \tag{2}$$

Opening proceeds are assumed retained at par, with no investment return. The original collateral returns on repayment and cancels against collateral held in the no-borrowing comparator. This is neither investor profit nor welfare. Actual LCU conversions, market spread, fee funding, interest payments and taxes are unobserved. Baseline assumptions are $r = 20\%$, $f = 0.5\%$, $s = 0.3\%$ per conversion and $g = \$40$; none is an estimated price for a deployed LCU protocol. The 20% rate is a round, explicitly hypothetical financing-cost scenario for comparison across FX paths; it is not a historical MakerDAO stability fee, an observed peso/lira loan quote or an estimate of an equilibrium LCU rate. There is no data-supported uniquely correct baseline, so the main results show the borrowing-rate grid alongside it.

The fee timing matters: charging the opening fee on discounted debt at closure would make its dollar value depend on the later FX rate and yield a different net series. We chose an externally paid opening fee and implement that same convention in every horizon and duration check. Likewise, USD 40 refers only to two specified borrower transactions; it is not an estimate of auction gas, oracle operation or all protocol transactions. Conversion friction in the liquidation branch is charged at entry only, because this stylised borrower makes no voluntary closing swap.

Table 2 Base scenario conventions. All amounts and rates are assumptions of the hypothetical LCU mapping, except the specified official FX series and observed MakerDAO draw amounts and dates. Changing any one convention requires rerunning the accompanying code.

Input	Base assumption and cash-flow interpretation
Annual borrowing rate	20% effective in the base case; sensitivity rates 5, 10, 40, 80 and 120%. Nominal LCU debt accrues as $(1+r)^t$, without intermediate payments.
Protocol fee	0.5% of original USD principal, paid separately at opening and not added to financed debt.
Conversion friction	0.3% on opening principal and 0.3% on the dollar value of accrued debt at voluntary closing.
Gas budget	USD 20 at opening in either branch and USD 20 for voluntary closing; no auction gas is modelled.
Loan duration	Selected clean-spell duration, or an imposed 3-, 6-, 12- or 24-month holding scenario.
Currency conversion	Daily official reference at selected observed spell endpoints; OECD/FRED monthly average for imposed horizons. Neither is a demonstrated executable borrower quote.
Collateral screen	150% terminal threshold and initial ratios of 150, 175 or 200%; no inferred liquidation probability.

4.2 Mutually exclusive liquidation

If the collateral is sold at time τ rather than voluntarily repaid, its USD mark before sale is $V_{i,\tau}$. A proportional auction haircut a gives $S_{i,\tau} = (1-a)V_{i,\tau}$. Under an explicit nonrecourse accounting convention, recovered debt is $R_{i,\tau} = \min(S_{i,\tau}, D_i(\tau))$, and a penalty $\lambda D_i(\tau)$ is collected *only from surplus*:

$$\Pi_{i,\tau} = \min\{\lambda D_i(\tau), [S_{i,\tau} - D_i(\tau)]_+\}, \tag{3}$$

$$H_{i,\tau} = [S_{i,\tau} - D_i(\tau)]_+ - \Pi_{i,\tau}, \qquad W_{i,\tau} = [D_i(\tau) - S_{i,\tau}]_+, \tag{4}$$

$$B_i^{\text{liq}}(\tau) = P_i - fP_i - sP_i - \tfrac{g}{2} - V_{i,\tau} + H_{i,\tau}. \tag{5}$$

H is cash returned to the borrower; W is the unrecovered loan claim. $V - H$ is the collateral value lost relative to holding it without borrowing. The borrower neither pays $D_i(\tau)$ again in cash nor repays $D_i(h)$ at the original horizon h. There is no voluntary exit swap or closing gas in this branch. The auction haircut and capped penalty are distinct sale effects; a different legal recourse rule would change who bears W.

The detailed numerical surplus and shortfall checks are in the accompanying cash-flow note and executable tests. The available collateral shocks do not reveal τ, oracle state, auction proceeds or haircut. Equation (5) is an implemented *conditional ledger*, not an estimated branch frequency or expected benefit. A deterministic terminal-ratio diagnostic is kept in the Appendix to show the boundary of the FX result without using it to modify the borrower medians.

5 Empirical design and robustness

Each draw is mapped to 3-, 6-, 12- and 24-month official-rate holding *scenarios*; these are not measured MakerDAO durations. Clean spells use observed start/end timestamps and $t = \text{days}/365.25$, with daily published FX dates at the endpoints in the primary short-spell calculation. Monthly marks are a like-for-like benchmark for the same clean sample, and the borrowing rate varies from 5% to 120% effective annually. We report event medians, positive shares, and $\sum_i B_i / \sum_i P_i$; the last is a *signed amount-weighted ratio*, not an average borrower's return. Means remain in machine-readable files but can be extremely negative for tiny draws facing fixed gas.

For dependence, we tabulate draws per urn, the chronologically first eligible draw from each urn, and an urn aggregate $100 \sum_{i \in u} B_i / \sum_{i \in u} P_i$. The pooled amount-weighted result is algebraically identical for all draws and urn aggregates; this is not evidence of independence. First-draw and equal-urn summaries answer different descriptive questions. To assess concentration, we rank events by *original principal size* before measuring what fraction of the full signed 12-month benefit is attributable to the largest 1, 5 and 10% of events. Month contributions use origination month. Finally, both 12-month and clean-duration mappings are rerun with minimum draw sizes of \$1, \$100, \$1,000, \$5,000 and \$10,000.

Table 3 Observed-duration counterfactual versus fixed holding-period scenarios. All entries are percent except clean-spell counts. The observed column uses official daily published reference dates for selected clean repaid spells; fixed columns use OECD/FRED monthly averages for all 130,742 eligible draw events per currency. Base assumptions: effective annual rate 20%, opening fee 0.5%, conversion 0.3% each way and gas $40 round trip. Gross excludes financing and execution costs.

FX	Clean spells	Observed net	12m gross	12m net	24m net
ARS	7,105	-2.86	24.61	2.67	25.58
TRY	7,105	-4.30	25.30	6.59	37.19

Table 4 Frequency and source check for the same 7,105 selected spells per currency at the 20% effective annual rate. Gross and net medians and positive net shares are percent. The same-source monthly mean averages the archived daily references in each calendar month; it differs in source and definition from OECD/FRED. Daily values use the most recent published date on or before each UTC endpoint. These are reference valuations, not executed swaps.

FX	Endpoint marks	*N*	Gross median	Net median	Net positive
ARS	OECD/FRED monthly	7,105	0.00	-3.00	13.03
ARS	Same-source monthly mean	7,105	0.00	-3.02	13.09
ARS	Official daily reference	7,105	0.62	-2.86	9.85
TRY	OECD/FRED monthly	7,105	0.00	-4.14	14.55
TRY	Same-source monthly mean	7,105	0.00	-4.16	14.64
TRY	Official daily reference	7,105	0.24	-4.30	15.57

The unit changes the question. An event median weights an urn with many draws many times; selecting the first eligible draw weights each represented urn once but discards subsequent activity. The median of within-urn amount-weighted ratios also assigns one observation to each urn while allowing its own large draws to receive greater weight. None of these three summaries identifies a representative person's payoff, because control of multiple urns by one person is unverified. The event mappings also share calendar-month FX observations. Accordingly, we report descriptive differences rather than conventional independent-event standard errors.

6 Results

6.1 Observed durations before long holding periods

The clean spell median is 6.95 days, and the hypothetical net median over *those durations* at daily reference marks is −2.86% for ARS and −4.30% for TRY (Table 3). Only 9.85% and 15.57% of mappings are net positive. The matched-source frequency comparison below makes the short-spell measurement limitation explicit. Selection for clean repayment precludes interpreting these results as a typical return or as an estimate of all-vault duration.

For 57.21% of selected spells, both endpoints fall in one calendar month, so either monthly method imposes *zero* gross FX movement on that subset. Daily marks show nonzero movement for 57.07% of those same-month ARS spells and 58.62% for TRY; some spells open and close on the same reference date. Relative to OECD/FRED monthly marks, the daily net median shifts from −3.00% to −2.86% in ARS and from −4.14% to −4.30% in TRY (Table 4). The matched-source monthly mean yields −3.02% and −4.16%, showing that source and aggregation both matter. The negative median survives, but the positive share changes, and the precise daily numbers should not be read as executable outcomes.

The round 20% financing rate does not drive the sign of the median alone: with 5% effective annual financing the daily medians are −2.10% and −3.36%, and with 40% they are −4.70% and −5.45% (Table 5). The cost of a particular peso or lira loan may lie outside this grid, especially if an official conversion cannot be accessed. At imposed 12-month horizons, changing the rate can reverse the sign; the full fixed-horizon grid is in `results/rate_sensitivity.csv`.

For comparison, with every draw timestamp artificially held open for 12 months under monthly OECD/FRED FX, gross median FX effects are 24.61% and 25.30%, while voluntary-repayment net medians are 2.67% and 6.59%. Holding for 24 months yields net medians of 25.58% and 37.19%. This chiefly contrasts *assumed horizons* with selected short spells; no typical borrower is shown to have captured the longer effect.

Table 5 Borrowing-rate sensitivity for the 7,105 selected observed-duration spells per currency using daily official reference marks. Effective annual rates and net benefit measures are percent; the fee, conversion and gas assumptions stay fixed. Rates are illustrative scenarios, not observed offers.

FX	Annual rate	Median net	Net positive
ARS	5%	-2.10	18.24
ARS	10%	-2.29	15.41
ARS	20%	-2.86	9.85
ARS	40%	-4.70	1.55
ARS	80%	-7.99	0.32
ARS	120%	-9.66	0.01
TRY	5%	-3.36	19.82
TRY	10%	-3.68	18.69
TRY	20%	-4.30	15.57
TRY	40%	-5.45	11.23
TRY	80%	-8.18	7.08
TRY	120%	-9.56	4.59

Table 6 Selected clean-spell duration bands at daily published FX marks. Net values are hypothetical LCU financing benefits evaluated at observed spell lengths. N is the number of repaid spells per currency band, not people. Positive is the percentage with positive conditional net benefit.

FX	Duration (days)	N	Median net	Positive
ARS	(0,1]	1,891	-9.09	0.00
ARS	(1,7]	1,676	-2.51	0.00
ARS	(7,30]	1,499	-2.41	3.34
ARS	(30,90]	1,026	-1.75	24.56
ARS	> 90	1,013	-2.73	39.29
TRY	(0,1]	1,891	-9.08	0.48
TRY	(1,7]	1,676	-2.86	6.92
TRY	(7,30]	1,499	-4.05	17.08
TRY	(30,90]	1,026	-3.46	36.74
TRY	> 90	1,013	-6.43	34.35

Table 7 Twelve-month hypothetical voluntary-repayment net benefit by observational unit (percent). Each urn's aggregate amount-weighted within-urn benefit is one urn unit. Weighted net pools signed benefit divided by original principal in the selected row. Rows reuse underlying draws and are not independent estimates.

FX	Unit	N	Median net	Mean net	Weighted net
ARS	All draw events	130,742	2.67	-11.85	12.64
ARS	First draw per urn	16,846	0.86	-21.56	10.21
ARS	Urn aggregate	16,846	1.86	-16.48	12.64
TRY	All draw events	130,742	6.59	-6.58	27.95
TRY	First draw per urn	16,846	-4.54	-23.39	25.48
TRY	Urn aggregate	16,846	-0.43	-14.34	27.95

6.2 Dependence among draws

Of 16,846 urns, 6,221 (36.93%) have one eligible draw, whereas 2,713 (16.10%) have more than ten and account for 70.29% of all draw events. The complete counts and band totals are in `results/urn_draw_counts.csv` and `results/urn_draw_distribution.csv`. Treating 130,742 events as independent borrowers would give misleading precision. The TRY event median is 6.59%, but the first-draw-per-urn median is -4.54% and urn-aggregate median is -0.43% (Table 7). ARS counterparts are 2.67%, 0.86% and 1.86%. Dependence and sample composition materially affect the interpretation, and urns need not represent distinct people. The contrast is substantively larger than a rounding change: even the sign of the TRY median depends on the unit used. It does not imply that the original event computation is arithmetically wrong; instead, a few heavily used vaults can contribute many event mappings. The draw-count distribution in the replication files lets readers see exactly how many events each urn contributes and repeat the comparison under another weighting rule.

Table 8 Effect of minimum draw size on hypothetical net benefits, under the same cost assumptions as Table 3. Observed rows use selected clean, repaid spell durations and daily FX marks; 12-month rows use the monthly OECD/FRED scenario. Weighted net is the sum of signed USD benefits divided by principal within each row.

FX	Sample	Min. USD	N	Median net	Weighted net
ARS	12m	1	130,742	2.67	12.64
ARS	12m	100	120,482	3.05	12.64
ARS	12m	1,000	95,125	4.86	12.65
ARS	12m	5,000	65,025	3.22	12.68
ARS	12m	10,000	50,710	3.14	12.71
ARS	Observed	1	7,105	-2.86	-0.80
ARS	Observed	100	5,795	-1.75	-0.79
ARS	Observed	1,000	3,970	-1.22	-0.77
ARS	Observed	5,000	2,642	-1.14	-0.77
ARS	Observed	10,000	2,073	-1.11	-0.77
TRY	12m	1	130,742	6.59	27.95
TRY	12m	100	120,482	10.94	27.95
TRY	12m	1,000	95,125	30.14	27.98
TRY	12m	5,000	65,025	32.37	28.06
TRY	12m	10,000	50,710	33.09	28.11
TRY	Observed	1	7,105	-4.30	0.21
TRY	Observed	100	5,795	-2.43	0.22
TRY	Observed	1,000	3,970	-1.34	0.23
TRY	Observed	5,000	2,642	-1.18	0.24
TRY	Observed	10,000	2,073	-1.16	0.24

Table 9 Concentration of hypothetical 12-month net benefit among the largest draw events, ranked by original principal. Share of signed net is the top group's sum of net USD benefit divided by the full sample's signed net USD benefit. All entries are percentages, not event probabilities.

FX	Largest draws	Share of principal	Share of signed net
ARS	1%	64.52	72.49
ARS	5%	83.41	88.03
ARS	10%	90.14	93.02
TRY	1%	64.52	66.85
TRY	5%	83.41	85.37
TRY	10%	90.14	91.69

6.3 Minimum size and contribution concentration

At the $1 floor, a fixed $40 voluntary gas budget disproportionately affects tiny positions: unweighted 12-month means are −11.85% (ARS) and −6.58% (TRY), whereas at a $100 threshold they become 5.82% and 12.74%. Small-position means are not typical returns. The five thresholds and exact denominators are in Table 8. Even above $10,000, the daily clean-duration net medians remain negative; their exact values are generated in the table and CSV. The largest 1% of draws represent 64.52% of original principal and 72.49% (ARS) or 66.85% (TRY) of signed 12-month net benefit. The largest 5% account for 88.03% and 85.37%. Origination timing also matters: October 2021 contributes 13.39% of full signed ARS net benefit, while May 2021 contributes 15.21% of TRY. These accounting shares do not show that very large borrowers could access uniform FX and gas prices without market impact.

The optional collateral exercise in the Appendix marks an assumed 12-month ratio, conditional on stipulated declines. Because no within-horizon collateral path, sale price or branch probability is observed, it cannot revise any net-benefit estimate in this section.

7 What the comparisons identify

The analysis contains three distinct forms of evidence. Draw timestamps, debt changes and urn identifiers come from the public MakerDAO record; the restricted clean repayment dates permit an observed spell length, subject to trace coverage and selection. Daily references and monthly official FX paths are published marks, not prices paid by these borrowers. All ARS/TRY loan balances, costs and net benefits are outcomes of a stipulated conversion and repayment rule. A positive 12-month counterfactual cannot establish that an actual borrower could remain open for 12 months, obtain official-rate ARS, or retain the original collateral throughout.

The negative clean-duration medians sharpen this distinction. At monthly frequency, a same-month opening and closing share one average and record no within-month FX movement despite charging all

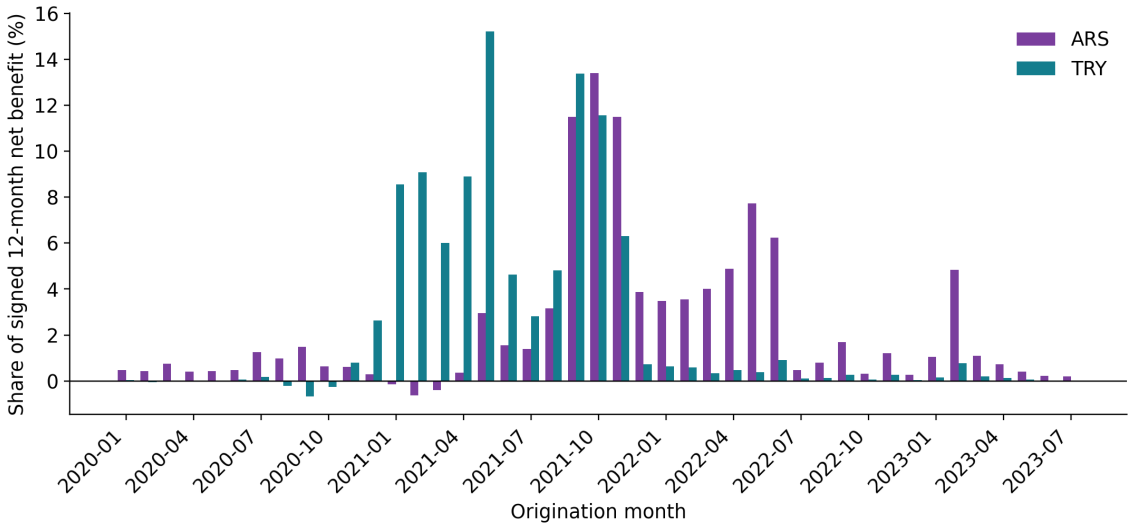


Fig. 3 Contribution by origination month to the signed amount-weighted 12-month net-benefit numerator. Shares can be negative. Complete principal weights and month results are in the accompanying CSV.

entry and closing costs; the daily check removes this artefact for spells spanning distinct published dates. Neither frequency demonstrates actual on-chain access to that rate. Nor does selecting successfully repaid, single-draw, non-liquidated spells establish that all ETH-A vaults had short lifetimes. An imposed long horizon is not an observed holding pattern.

The cost, size and concentration checks describe where the accounting result comes from. Raising a size floor changes both the fixed gas share of principal and the composition of months and urns; it does not merely remove numerical outliers. Similarly, the fraction of a *signed* aggregate attributable to the largest draws can be high even if the median event is modest. For executable gains, large trades could move the relevant market price. The current data contain neither such trading costs nor a parallel ARS quote, so we stop at a transparent conditional comparison.

8 Accounting boundary for protocol solvency

The borrower benefit is not a measure of issuer capital. Borrower-posted collateral is encumbered until enforcement and is not a free asset in addition to its loan claim. Let A_t^{loan} be performing LCU loan claims, A_t^{recovery} cash after completed collateral sales with the related claim removed, R_t unrestricted reserves, and Q_t^{redeem} and Q_t^{other} token redemption and other obligations. Then a schematic equity identity is

$$K_t = A_t^{\text{loan}} + A_t^{\text{recovery}} + R_t - Q_t^{\text{redeem}} - Q_t^{\text{other}}. \tag{6}$$

A completed sale replaces the claim with recovery; $W_{i,\tau} = D_i(\tau) - R_{i,\tau}$ writes down an *asset*, while token burns or settlement must separately reduce liabilities. We have no observed LCU liabilities, unrestricted reserves or auction proceeds from which to compute K_t. This identity only identifies data needed for a later issuer-solvency study; reserve, oracle and auction design proposals require independent testing [5, 8, 14].

9 Limitations and interpretation

Published daily and monthly FX references are not necessarily executable, especially in ARS; an unknown parallel premium has no known bias direction. A same-date published reference may have arrived after the transaction. Clean spells are selected and the quote calendar is less granular than an intraday execution record. Repeated draws share urn-level risks and calendar FX shocks; no causal, confidence-interval or independent-borrower claim follows. Fixed horizons are hypothetical and cannot be inferred from clean duration. Fees, gas, conversion friction and interest rates are scenarios, not observed transaction prices for a hypothetical LCU protocol. Large positions omit market impact. The liquidation ledger assumes nonrecourse, while within-horizon collateral prices, oracles, auction discounts and settlement times are unavailable. Terminal ratios cannot identify expected cash flows. Peg stability, reserve adequacy, governance backstops, taxes and legal exchange access remain outside this dataset.

10 Conclusion

The most directly observed duration fact is that selected clean, repaid MakerDAO spells last a median 6.95 days. At hypothetical LCU debt terms and daily published FX reference marks, median net effects over those observed lengths are −2.86% in ARS and −4.30% in TRY at the illustrative 20% effective rate. They remain negative at 5% and 40% rates; the monthly-frequency check also yields negative medians, though it misses within-month FX movement. Positive medians at 12 and 24 months apply to *imposed holding periods*; amount-weighted aggregates are highly concentrated among large draws and particular months. Daily official marks are not demonstrated executable prices. The collateral ledger and issuer identity set accounting boundaries; they do not establish liquidation probabilities, a deployed protocol or its solvency.

Appendix: deterministic terminal collateral mark

This auxiliary screen is separated from the empirical borrower comparisons because it has no asset-specific historical collateral paths or sale data. At an assumed 12-month endpoint, initial collateral worth $c_0 P_i$ loses a stipulated fraction q, giving

$$\mathrm{CR}_{i,12}^{\mathrm{terminal}}(q, c_0) = \frac{c_0 P_i (1-q)}{D_i(1)}, \qquad I_{i,12}^{\mathrm{terminal}} = \mathbf{1}\{\mathrm{CR}_{i,12}^{\mathrm{terminal}} < 1.50\}. \tag{7}$$

A breach share is the percentage of *event mappings* below an assumed 150% threshold at month 12, not a probability of earlier liquidation or auction completion. The generic 0, 10, 25, 40 and 60% declines are stipulated without shock probabilities. ETH and BTC are not distinguished because the same percentage shocks would give identical results. For a 175% initial ratio and 25% terminal decline, the screen flags 59.92% of ARS and 53.41% of TRY mappings; no expected-loss or risk-adjusted return is calculated.

Table 10 Auxiliary terminal-only screen for a 175% initial ratio, 150% threshold, 20% effective annual rate and generic stipulated declines. Percentages refer to 130,742 event mappings per currency at month 12, not liquidation probabilities.

FX	Price decline	Terminal breach share
ARS	10%	0.00
ARS	25%	59.92
ARS	40%	94.48
ARS	60%	98.87
TRY	10%	3.76
TRY	25%	53.41
TRY	40%	58.29
TRY	60%	100.00

Data and code availability

The public reproducibility package contains processed event and lifecycle files, archived monthly and daily official FX inputs, source query and checksums, executable borrower cash-flow model, tests, generated result tables, figures and build instructions. The repository is https://github.com/nikorokni/inflation-driven-debt-erosion-defi. The original public MakerDAO archive is linked with a checksum in the package rather than duplicated.

Declarations

Conflict of interest. No conflict of interest was reported.
Funding. No external funding was reported.
Ethics. The analysis uses public blockchain traces and published aggregate exchange rates.

References

[1] Schär F. Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*. 2021;103(2):153–174. doi:10.20955/r.103.153-74.

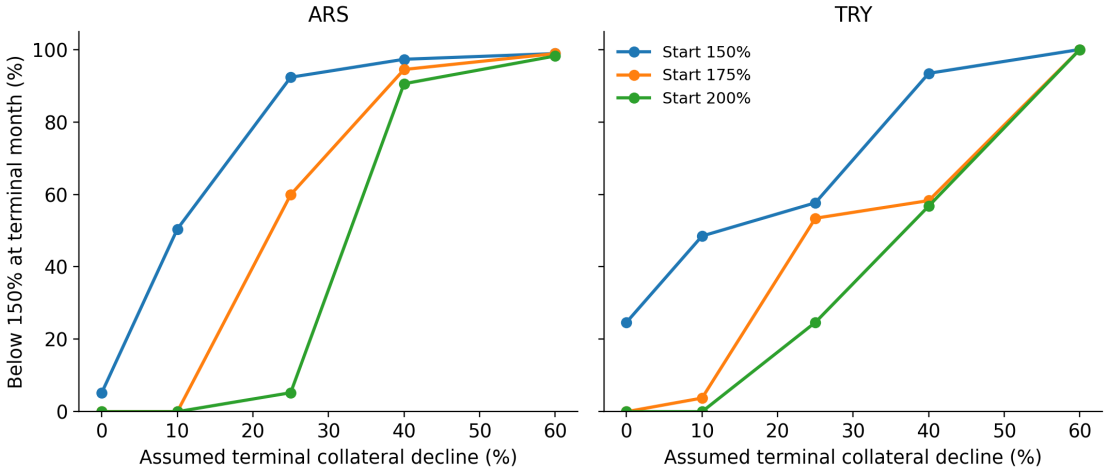


Fig. 4 Terminal ratio screen over deterministic collateral declines and initial ratios; no asset-specific price process or executed auction is estimated.

[2] Werner SM, Perez D, Gudgeon L, et al. SoK: Decentralized finance (DeFi). In: *Proceedings of the 4th ACM Conference on Advances in Financial Technologies*. 2022. pp. 30–46. doi:10.1145/3558535.3559780.

[3] Lyons RK, Viswanath-Natraj G. What keeps stablecoins stable? *Journal of International Money and Finance*. 2023;131:102777. doi:10.1016/j.jimonfin.2022.102777.

[4] Klages-Mundt A, Minca A. While stability lasts: A stochastic model of noncustodial stablecoins. *Mathematical Finance*. 2022;32(4):943–981. doi:10.1111/mafi.12357.

[5] Gorton GB, Zhang JY. Taming wildcat stablecoins. *University of Chicago Law Review*. 2023;90(3):909–971.

[6] Chaleenutthawut Y, Davydov V, Evdokimov M, et al. Loan portfolio dataset from MakerDAO blockchain project. *IEEE Access*. 2024;12:24843–24854. doi:10.1109/ACCESS.2024.3363225.

[7] Sky Protocol Docs. Vat (Core Accounting). *Sky Protocol Documentation*. https://developers.skyeco.com/protocol/core/vat/ (accessed 28 September 2026).

[8] Consultative Council of Directors of Financial Stability. Financial stability risks from cryptoassets in emerging market economies. *BIS Papers*. 2023;(138). https://www.bis.org/publ/bppdf/bispap138.pdf.

[9] Organisation for Economic Co-operation and Development. National currency/USD, average daily rates: Argentina [ARGCCUSMA02STM]. *FRED, Federal Reserve Bank of St. Louis*. 2026. https://fred.stlouisfed.org/series/ARGCCUSMA02STM (accessed 8 August 2026).

[10] Organisation for Economic Co-operation and Development. National currency/USD, average daily rates: Türkiye [CCUSMA02TRM618N]. *FRED, Federal Reserve Bank of St. Louis*. 2026. https://fred.stlouisfed.org/series/CCUSMA02TRM618N (accessed 8 August 2026).

[11] Banco Central de la República Argentina. Tipo de Cambio de Referencia: Comunicación A3500 (mayorista), historical daily workbook. *Estadísticas del BCRA*. https://www.bcra.gob.ar/archivos/Pdfs/PublicacionesEstadisticas/com3500.xls (accessed 12 October 2026).

[12] European Central Bank. Euro foreign exchange reference rates: historical time series, USD and TRY per EUR. *ECB Data*. https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip (accessed 12 October 2026).

[13] Qin K, Zhou L, Gamito P, Jovanovic P, Gervais A. An empirical study of DeFi liquidations: Incentives, risks, and instabilities. In: *Proceedings of the 21st ACM Internet Measurement Conference*. 2021:336–350. doi:10.1145/3487552.3487811.

[14] Arner DW, Auer R, Frost J. Stablecoins: Risks, potential and regulation. *BIS Working Papers*. 2020;(905). https://www.bis.org/publ/work905.htm.